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CP372 Assignment 2 Report
DS-FTP (Stop-and-Wait and Go-Back-N)

DS-FTP Control Packets (SOT & EOT)

Start of Transmission (SOT)

- The SOT packet is used to start a transfer session between the sender and receiver.
- It is sent by the sender first to begin the connection.
- SOT always has:
    - Type = 0
    - SeqNum = 0
    - Length = 0
- When the receiver gets the SOT packet:
    - It sends back an ACK with sequence number 0.
    - This confirms that both sides are synced and ready to send data.
- Even though it's just a control message, SOT still follows the standard DS-FTP packet format:
    - Total packet size is 128 bytes.
    - Fields include Type, SeqNum, Length, and Payload.
    - The payload exists but is ignored for control packets.

End of Transmission (EOT)

- The EOT packet is used to end the transfer session.
- It is sent after the data transfer phase is complete.
- EOT uses:
    - Type = 3
    - Length = 0
    - SeqNum = (last DATA sequence number + 1) mod 128
- This keeps the sequence numbers consistent across the transfer.
- When the receiver gets the EOT packet:
    - It sends an ACK for that sequence number.
    - The connection then closes cleanly.

Protocol Design

- Both SOT and EOT use the same 128-byte DS-FTP packet structure:
    - Type
    - SeqNum
    - Length
    - Payload
- Keeping the same structure makes the protocol consistent, simple, and easy to process.

Protocol	Window	FileSize	RN	Run1(s)	Run2(s)	Run3(s)	Average(s)	SuccessRuns
Stop-and-Wait	N/A	Small(<4KB)	0	0.04	0.04	0.04	0.04	3
Stop-and-Wait	N/A	Small(<4KB)	5	1.3	1.31	1.32	1.31	3
Stop-and-Wait	N/A	Small(<4KB)	100	0.06	0.05	0.04	0.05	3
Stop-and-Wait	N/A	Large(0.25MB)	0	0.89	0.93	1.04	0.95	3
Stop-and-Wait	N/A	Large(0.25MB)	5	170.55	170.76	171.71	171.01	3
Stop-and-Wait	N/A	Large(0.25MB)	100	8.2	8.59	8.21	8.33	3
GBN-20	20	Small(<4KB)	0	0.06	0.05	0.05	0.05	3
GBN-20	20	Small(<4KB)	5	0.05	0.06	0.06	0.06	3
GBN-20	20	Small(<4KB)	100	0.05	0.05	0.05	0.05	3
GBN-20	20	Large(0.25MB)	0	0.81	0.75	0.89	0.82	3
GBN-20	20	Large(0.25MB)	5	0.71	0.64	0.76	0.7	3
GBN-20	20	Large(0.25MB)	100	2.83	4.11	4.32	3.75	3
GBN-40	40	Small(<4KB)	0	0.15	0.15	0.22	0.17	3
GBN-40	40	Small(<4KB)	5	0.21	0.21	0.19	0.2	3
GBN-40	40	Small(<4KB)	100	0.17	0.16	0.16	0.16	3
GBN-40	40	Large(0.25MB)	0	5.05	0.78	0.8	2.21	3
GBN-40	40	Large(0.25MB)	5	0.72	0.84	0.8	0.79	3
GBN-40	40	Large(0.25MB)	100	0.8	0.73	0.75	0.76	3
GBN-80	80	Small(<4KB)	0	0.06	0.07	0.05	0.06	3
GBN-80	80	Small(<4KB)	5	0.05	0.08	0.06	0.06	3
GBN-80	80	Small(<4KB)	100	0.06	0.06	0.06	0.06	3
GBN-80	80	Large(0.25MB)	0	0.69	0.67	0.79	0.72	3
GBN-80	80	Large(0.25MB)	5	0.68	0.65	0.7	0.68	3
GBN-80	80	Large(0.25MB)	100	0.64	0.67	0.66	0.66	3

- Each setup in the performance table was run three times, and the final result is the average of those runs:
- Average = (Run1 + Run2 + Run3) / 3
- All tests finished successfully (SuccessRuns = 3), so the averages are based on valid transfers.
- From the results, Go-Back-N is usually faster than Stop-and-Wait, especially with large files or ACK loss. For large files, GBN-80 performed the best (~0.66–0.72 s), while Stop-and-Wait slowed down a lot when ACK loss happened (up to 171.01 s). For small files, GBN stayed very fast (~0.05–0.20 s), while Stop-and-Wait increased to about 1.31 s under higher loss.